

Lesson 5: Equally Likely Outcomes and Counting

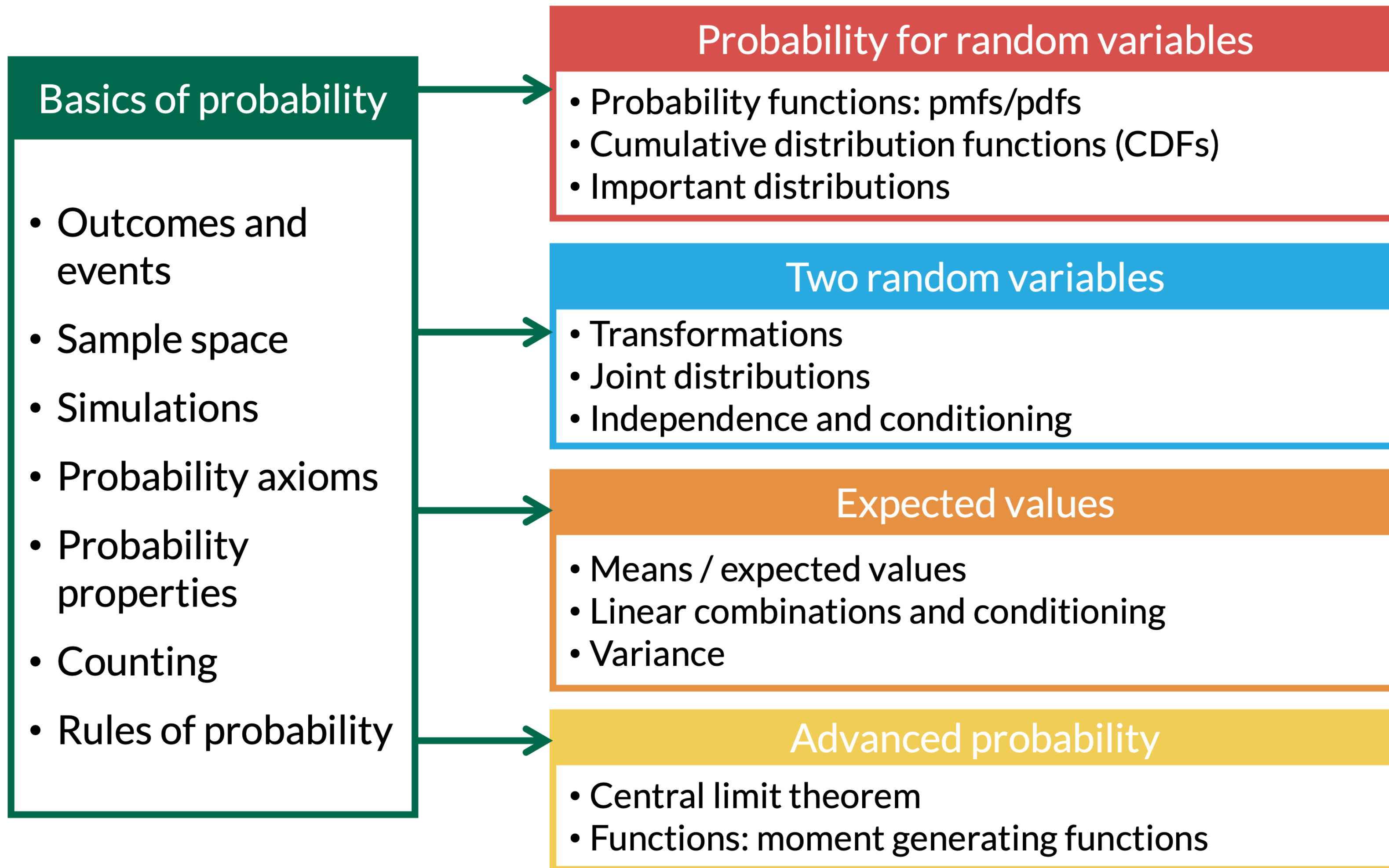
Meike Niederhausen and Nicky Wakim

2026-10-12

Learning Objectives

1. Define **permutations** and **combinations**
2. Calculate the probability of sampling any combination of the following: *with or without replacement* and *order does or does not matter*
3. Use a simulation to check a probability found by counting

Where are we?



Birthday example



Welcome to the party! Today you will participate in an **interactive experiment** about the birthday paradox. We will use you, the reader, as part of our data, to help explain what it is, why it is cool, and how it works.

By [Russell Samora](#)

Let's do this!

Basic Counting Examples (1/3)

Example 1

Suppose we have 10 (distinguishable) subjects for study.

1. How many possible ways are there to order them?
2. How many ways to order them if we can reuse the same subject and
 - need 10 total?
 - need 6 total?
3. How many ways to order them *without replacement* and only need 6?
4. How many ways to choose 6 subjects without replacement if the order doesn't matter?

Basic Counting Examples (2/3)

Suppose we have 10 (distinguishable) subjects for study.

Example 1.1

How many possible ways are there to order them?

Example 1.2

How many ways to order them if we can reuse the same subject and

- need 10 total?
- need 6 total?

Basic Counting Examples (3/3)

Suppose we have 10 (distinguishable) subjects for study.

Example 1.3

How many ways to order them without replacement and only need 6?

Example 1.4

How many ways to choose 6 subjects without replacement if the order doesn't matter?

Learning Objectives

1. Define permutations and combinations

2. Calculate the probability of sampling any combination of the following: *with or without replacement* and *order does or does not matter*

3. Use a simulation to check a probability found by counting

Permutations and Combinations

Definition: Permutations

Permutations are the number of ways to **arrange in order** r distinct objects when there are n total.

$${}_nP_r = \frac{n!}{(n-r)!}$$

Definition: Combinations

Combinations are the number of ways to choose (**order doesn't matter**) r objects from n without replacement.

$${}_nC_r = \text{"n choose r"} = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

- We will see combinations a lot in the binomial distribution

Some combinations properties

Property

Proof

$$\binom{n}{r} = \binom{n}{n-r} \qquad \binom{n}{r} = \frac{n!}{r!(n-r)!} \text{ and } \binom{n}{n-r} = \frac{n!}{(n-r)!(n-(n-r))!} = \frac{n!}{(n-r)!r!}$$

$$\binom{n}{1} = n \qquad \binom{n}{1} = \frac{n!}{1!(n-1)!} = \frac{n \cdot (n-1) \cdot (n-2) \cdots 1}{1! \cdot (n-1) \cdot (n-2) \cdots 1} = \frac{n \cdot (n-1)!}{1 \cdot (n-1)!} = \frac{n}{1} = n$$

$$\binom{n}{0} = 1 \qquad \binom{n}{0} = \frac{n!}{0!(n-0)!} = \frac{n!}{1 \cdot n!} = 1$$

Learning Objectives

1. Define **permutations** and **combinations**

2. Calculate the probability of sampling any combination of the following: *with or without replacement* and *order does or does not matter*

3. Use a simulation to check a probability found by counting

Table of different cases

- n = total number of objects
- r = number objects needed

	with replacement	without replacement
order matters	n^r	$nPr = \frac{n!}{(n-r)!}$ (permutations)
order doesn't matter	$\binom{n+r-1}{r}$	$nCr = \binom{n}{r} = \frac{n!}{r!(n-r)!}$ (combinations)

Enumerating Events and Sample Space

- Recall, $P(A) = \frac{|A|}{|S|}$
 - Within combinatorics, we can use the previous equations to help enumerate the event and sample space
 - But A might be a combination of enumerations
- For example in the following example drawing 2 spades when order does not matter, we actually need to enumerate the other cards that are NOT spades. So the event is choosing 2 spades out of 13 AND choosing 0 other cards of 39 cards (13 hearts + 13 clubs + 13 diamonds).
- Thus the probability is actually:

$$P(\text{two spades}) = \frac{\binom{13}{2} \binom{39}{0}}{\binom{52}{2}}$$

- Note that $13 + 39 = 52$ and $2 + 0 = 2$. So the numerator's n 's add up to the denominator's n and the numerator's r 's add up to the denominator's r 's

Example 2

Suppose we draw 2 cards from a standard deck without replacement. What is the probability that both are spades when

1. order matters?
2. order doesn't matter?

Example 2

Suppose we draw 2 cards from a standard deck without replacement. What is the probability that both are spades when

1. order matters?
2. order doesn't matter?

We can do a simulation! First, a tiny version with 5 simulations:

```
1 set.seed(1234)
2 suits <- c("S", "H", "C", "D")
3 deck <- rep(suits, each = 13)
4
5 reps <- 5
6 hands <- expand_grid(sim = 1:reps, draw = 1:2) |>
7   group_by(sim) |>
8   mutate(card = sample(deck, n(), replace = FALSE)) |>
9   ungroup()
10 hands
```

```
# A tibble: 10 × 3
   sim draw card
  <int> <int> <chr>
1     1     1 C
2     1     2 H
3     2     1 H
4     2     2 C
5     3     1 D
6     3     2 D
7     4     1 S
8     4     2 S
9     5     1 C
10    5     2 H
```

Example 2

Suppose we draw 2 cards from a standard deck without replacement. What is the probability that both are spades when

1. order matters?
2. order doesn't matter?

Same code, now with 100,000 simulations. Then check each simulated hand:

```
1 set.seed(1234)
2 reps <- 100000
3 hands <- expand_grid(sim = 1:reps, draw = 1:2) |>
4   group_by(sim) |>
5   mutate(card = sample(deck, n(), replace = FALSE)) |>
6   ungroup()
7
8 spade_results <- hands |>
9   group_by(sim) |>
10  summarise(ordered = card[1] == "S" & card[2] == "S",
11            unordered = sum(card == "S") == 2)
12
13 spade_results |>
14   summarise(prob_ordered = mean(ordered),
15             prob_unordered = mean(unordered))
```

```
# A tibble: 1 × 2
  prob_ordered prob_unordered
      <dbl>         <dbl>
1    0.0588         0.0588
```

Exact answer: $\text{choose}(13, 2) / \text{choose}(52, 2) = 0.0588$

Learning Objectives

1. Define **permutations** and **combinations**
2. Calculate the probability of sampling any combination of the following: *with or without replacement* and *order does or does not matter*
3. Use a simulation to check a probability found by counting

Birthday paradox revisited (1/3)

Example 3

Suppose 23 people are in a room, and every day of the year is equally likely to be a birthday (ignore leap days). What is the probability that at least two people share a birthday?

```
1 set.seed(550)
2 reps <- 100000
3 n_people <- 23
4 birthdays <- expand_grid(sim = 1:reps, person = 1:n_people) |>
5   mutate(birthday = sample(1:365, n(), replace = TRUE))
6 head(birthdays, 4)
```

```
# A tibble: 4 × 3
  sim person birthday
<int> <int>    <int>
1     1     1     145
2     1     2     246
3     1     3     284
4     1     4      90
```

Birthday paradox revisited (2/3)

```
1 room_results <- birthdays |>  
2   group_by(sim) |>  
3   summarise(n_different = n_distinct(birthday)) |>  
4   mutate(shared = n_different < n_people)  
5  
6 room_results |>  
7   summarise(prob_shared = mean(shared))
```

①

②

③

④

```
# A tibble: 1 × 1  
  prob_shared  
    <dbl>  
1      0.508
```

Exact answer:

$$1 - \frac{365 \cdot 364 \cdots 343}{365^{23}} = 0.5073$$

Birthday paradox revisited (3/3)

Sensitivity analysis: what if there are 30 people in the room?

```
1 reps <- 100000
2 n_people <- 30
3 birthdays <- expand_grid(sim = 1:reps, person = 1:n_people) |>
4   mutate(birthday = sample(1:365, n(), replace = TRUE))
5
6 birthdays |>
7   group_by(sim) |>
8   summarise(n_different = n_distinct(birthday)) |>
9   mutate(shared = n_different < n_people) |>
10  summarise(prob_shared = mean(shared))
```

```
# A tibble: 1 × 1
  prob_shared
    <dbl>
1      0.707
```

①

Learning Objectives

1. Define **permutations** and **combinations**
2. Calculate the probability of sampling any combination of the following: *with or without replacement* and *order does or does not matter*
3. Use a simulation to check a probability found by counting